Ram Moment Coordinates and Certified Posterior-Law Refinements for Three-State Reconstruction

M439, Family 229: Source Audit and Local Improvements

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Abstract

We examine the pinned September 25, 2026 manuscript on the exact reconstruction threshold for the three-state symmetric channel. We distinguish the analytic positive-channel proof, the negative-channel degree-28 pair certificates, the weighted-logarithm interval calculation, and the 469-orbit finite experiment with grid denominator 72. We specify the posterior law, product likelihood tilt, boundary conventions, and dependency on each unexecuted certificate. Actual private Ram passages supply exact finite polynomial moment coordinates and a disciplined separation of law, representation, and enclosure error, not a replacement reconstruction theorem. We prove a sharp pointwise formula for the second-moment increase caused by the source's barycentric spread, and construct a stratum-aware integer completion that is Blackwell-between the original finite law and the source's pure completion for the same input. The latter has an exact improvement ledger, preserves the fixed integer mass denominator, and extends to symmetric laws on any finite alphabet. Bounded rational checks are separated from the source's large certificate runs. No global threshold, stochastic-block-model theorem, historical novelty, or runtime improvement is independently claimed. Contribution assessment: C3/5, an exact law-preserving completion improvement and sharp spread budget are proved with finite rational witnesses while the global certificate runs and full-recursion performance remain unverified.

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1 Pinned Problem and Evidence Boundary

1.1 Pin, quantifiers, and target

The source is OpenAI, *The exact reconstruction threshold for the three-state symmetric channel*, September 25, 2026, at catalogue commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The catalogue row is M439, family 229, applicability L5. Applicability is not a contribution rating. Relative to the study root, the source directory is

`_sources/openai-math/preprints/The-exact-reconstruction-threshold-for-the-three-state-symmetric-channel-September-25-2026`.

All seven substantive sections and all four mathematical appendices have been read, including the mathematical interpretation of the code and the source's recorded executions. The report audits those dependencies and proves the local results below; reading is not an independent verification of every source theorem.

For every integer $b \geq 2$, or every real $d > 1$ with observed Poisson Galton–Watson offspring, the root spin is uniform on $\{0, 1, 2\}$. All tree variables are independent of the root spin. Conditional on a parent, independent edges transmit through

$$K_\lambda(i, j) = \lambda \mathbf{1}_{i=j} + (1 - \lambda)/3, \qquad -\tfrac{1}{2} \leq \lambda \leq 1. \tag{1.1}$$

In the regular model $D \equiv b$ and $d = b$; in the Poisson model $D \sim \mathrm{Pois}(d)$. The observation is the rooted, unlabelled tree through depth n, together with the spin attached to each depth-n vertex. Interior spins and sibling orders are not observed. Let

$$a_n = \mathbb{E} \left\| \mathbb{P}(S_\rho = \cdot \mid T_{\leq n}, S_{\partial T_n}) - \boldsymbol{u} \right\|_{\mathrm{TV}}, \qquad \boldsymbol{u} = (1/3, 1/3, 1/3).$$

The expectation includes tree and spins, *without conditioning on survival*. The pinned assertion is reconstruction if and only if $d\lambda^2 > 1$, with non-reconstruction at admissible equality for both signs. Negative equality is possible only at $d \geq 4$, including coloring at $(d, \lambda) = (4, -1/2)$. The cases $\lambda = 0$ and an empty generation are uninformative. The case $\lambda = 1$ is above threshold for these tree families.

The source also states an SBM consequence with independent uniform community labels, fixed positive within/between rates a, b, $d = (a+2b)/3 > 1$, and overlap maximized over label permutations. Its boundary is $(a - b)^2 = 3(a + 2b)$. The tree-to-graph impossibility transfer and the $O(n \log n)$ achievability algorithm are external dependencies, not algorithms implemented here. We reserve $a = -\lambda$ for negative channel magnitude outside this SBM paragraph.

1.2 Claim classes

Class	Scope in this report
Inherited source claim	The full threshold, uniform negative fixed-point gap, all fourteen large pair polynomials, interval completion, and the 28 grid stopping runs.
Locally proved	Calibration and boundary facts used here; exact spread inflation; capped residual completion; fixed-denominator implementation; finite-alphabet extension; exact moment-coordinate realization.
Bounded execution	Rational identity and finite-law checks recorded in the author check receipt; these do not stand in for the source's 37-record replay.
Conditional consequence	Any valid upper recursion using the new completion can be used with the source's fixed-point gap; a successful stopping certificate and the gap remain required.
Not established	A shorter 469-orbit run, fewer interval boxes, independently verified full threshold, arbitrary-offspring critical theorem, or new historical priority.

2 Exact Posterior Laws and Bayes Products

2.1 Spaces, invariants, and law normalization

All observation spaces are finite or standard Borel. Put

$$\Delta_2 = \{p \in [0, 1]^3 : \textstyle\sum_i p_i = 1\}, \quad m = 3p, \quad v = m - 1, \quad \mathcal{M} = 3\Delta_2, \quad \omega = e^{2\pi \mathrm{i}/3}.$$

For a probability law Q on Δ_2 to be a canonical posterior law under uniform prior, it is necessary and sufficient that $\int p_i \, \mathrm{d}Q = 1/3$. Indeed the joint law

$$\mathbb{P}(S = i, P \in \mathrm{d}p) = p_i Q(\mathrm{d}p) \tag{2.1}$$

has that prior and posterior P. Equivalently the conditional law is $3p_iQ(\mathrm{d}p)$. Coordinate-permutation invariance implies balance. It does not mean that only the permutation orbit is observed: the experiment observes the *labelled* vector. An orbit representation samples an orbit and a uniform element of the six-element permutation group, retaining the resulting vector; repeated coordinates are counted with their stabilizer multiplicity.

Set

$$\zeta = \sum_{i=0}^{2} p_i \omega^i, \quad x = |\zeta|^2 = \frac{3\sum_i p_i^2 - 1}{2}, \quad y = \Re\zeta^3 = \frac{1}{2}\prod_i v_i. \tag{2.2}$$

Then $\langle v^2 \rangle = 2x$, $\langle v^3 \rangle = 2y$, $v_i^3 = 3xv_i + 2y$, and

$$0 \le x \le 1, \quad |y| \le x^{3/2}, \quad 1 - 3x + 2y \ge 0. \tag{2.3}$$

For $x > 0$, $r = y/x$ satisfies $-1/2 \le r \le 1$ and $r^2 \le x \le (3 - 2r)^{-1}$. These are necessary actual-simplex constraints, not permission to assign arbitrary independent moments. The covariance matrix of a few invariants is likewise insufficient to realize a law.

For $\mu_n = \mathbb{E}x(P_n)$,

$$\mu_n/\sqrt{3} \le a_n \le (2/3)\sqrt{\mu_n}. \tag{2.4}$$

To check constants, for three zero-sum coordinates v_i, $\mathrm{TV}(p, \boldsymbol{u}) = \max_i |v_i|/3$. Their squared sum is at most $2(\max_i |v_i|)^2$, while $|v_i| \le 2|\zeta|$. Hence $|\zeta|/\sqrt{3} \le \mathrm{TV} \le 2|\zeta|/3$. Use $|\zeta|^2 \le |\zeta|$ and Cauchy–Schwarz. These constants use the source's normalization, not an unnormalized χ^2 convention.

2.2 The product measure is not the output measure

An edge maps m to $1 + \lambda(m-1)$, and thus (x, y) to $(\lambda^2 x, \lambda^3 y)$. For k independent *marginal* child messages, let

$$W_i = \prod_{j=1}^{k}(1 + \lambda v_{i,j}), \qquad Z = \tfrac{1}{3}\sum_i W_i, \qquad p_i^{\mathrm{out}} = W_i/(3Z). \tag{2.5}$$

Their true combined marginal law is the product law tilted by Z. Thus

$$\int f \, \mathrm{d}\mathcal{R}_{\lambda,D}(Q) = \sum_{k \ge 0} \mathbb{P}(D = k) \int Z f(p^{\mathrm{out}}) \, \mathrm{d}Q^{\otimes k}. \tag{2.6}$$

Since $\mathbb{E}(1 + \lambda v_i) = 1$, each fixed-count integral has mass one. At $k = 0$ take $Z = 1$, $p^{\mathrm{out}} = \boldsymbol{u}$. At $Z = 0$ take an arbitrary output, say $\boldsymbol{u}$, and set $Zf = 0$. This set has zero tilted probability. For bounded continuous f, this extension is continuous: $|Zf| \le \|f\|_\infty Z \to 0$. A bound on $1/Z$ alone would miss this cancellation, especially at coloring.

A concrete guardrail from the independent audit uses two plain inputs, each uniform over permutations of $(1/2, 1/2, 0)$. Coincident zeros have probability $1/3$, tilt $3/2$, and output $x = 1/4$; different zeros have probability $2/3$, tilt $3/4$, and output $x = 1$. The correct moment is $5/8$, whereas unweighted averaging gives $3/4$. Thus even exact arithmetic on a grid cannot repair an incorrectly specified posterior law.

2.3 Limits, extinction, and what degradation proves

If $\mathcal{E} \preceq \mathcal{F}$ denotes root-independent garbling, posterior sufficiency and conditional Jensen give $\mathbb{E}x(\mathcal{E}) \le \mathbb{E}x(\mathcal{F})$ and the analogous inequality for x^2. Branch composition, edge transmission, and mixtures with observed root-independent counts preserve this order. The source uses the decreasing fields generated by *all* depth observations from n onward, not the individual depth fields, to obtain a reverse-martingale posterior limit. Formula (2.6) is weakly continuous: truncate the offspring count, use the preceding zero-normalizer extension, then bound the tail by $\|f\|_\infty \mathbb{P}(D > K)$. The limiting law is a symmetric fixed point.

Writing s for survival probability ($s = 1$ regularly), extinction gives $x_\infty = 0$ and

$$\mu^2 \le s\nu, \quad |\eta| \le \sqrt{\mu\nu}, \qquad \nu = \mathbb{E}x^2, \quad \eta = \mathbb{E}y.$$

For Poisson offspring $-\log(1-s) = ds$ and $s \le 2(1-1/d)$. Conditioning the recursion on survival would change its independent-offspring structure; no such step is used.

Decreasing λ by a factor $c \in [0,1]$ at fixed sign is a garbling using independent group increments along the observed tree, with shared path increments retained. Thinning Poisson children decreases d by a garbling. These statements compare entire experiments, not just marginal boundary labels. They explain how closed parameter rectangles can be certified from their upper corners.

3 Stepwise Audit of the Pinned Proof

3.1 Positive equality and the radial dependency

For strictly positive m, natural logarithms, and $\langle h \rangle = \frac{1}{3}\sum_i h_i$, the source defines

$$I = \langle m \log m \rangle, \quad J = \tfrac{1}{2}\langle v \log m \rangle, \quad u_i = \log m_i - \langle \log m \rangle, \quad B = \tfrac{1}{2}\langle m u^2 \rangle, \quad F = 3I - 2J + \tfrac{4}{5}B.$$

At an interior-channel fixed point the logarithms are integrable: edge coordinates lie in $[c_-, c_+]$ with $c_- > 0$, and a k-branch output has $|\log m_i^{\mathrm{out}}| \le k \log(c_+/c_-)$. Regular and Poisson counts have all required moments. Conditional independence gives

$$\mathbb{E}J = d\mathbb{E}J_e, \qquad \mathbb{E}B = d\mathbb{E}B_e + 2\alpha(\mathbb{E}J)^2, \quad \alpha = \mathbb{E}[D(D-1)]/d^2.$$

With H the relative entropy of the branch likelihood tilt, the identity is

$$\mathbb{E}(F - dF_e) = -3H + \tfrac{8}{5}\alpha(\mathbb{E}J)^2. \tag{3.1}$$

The source's radial appendix proves, for $0 < \lambda < 1$,

$$F(m) - \lambda^{-2}F(1+\lambda v) \ge \tfrac{1}{2}(1-\lambda^2)x^2.$$

Its critical sign step is not a grid evaluation. With $\partial = \tau\frac{d}{d\tau}$ on a radial ray, it proves $\partial(\partial-1)(\partial-2)F \ge 12x^2$, using $D_t = 1 - 3t^2x + 2t^3y > 0$ and

$$D^3(G - 12x^2) = R + \int_0^1 N(t)/D_t \,\mathrm{d}t, \qquad D_t^{-1} \ge 2 - D_t.$$

The affine $N(t)$ is nonnegative at both endpoints; the remaining polynomial $L = R + \int N(t)(2-D_t)\,\mathrm{d}t$ is nonnegative. The appendix makes the substitutions $y = rx$, $x = (r^2 + z/(3-2r))/(1+z)$ and $r = (p-1/2)/(1+p)$, exposing nonnegative factors plus positive quadratic tails. The three quadratic discriminants are -7189911, -34724016, and -48247452. These are algebraic certificates

on full domains; their bounded identities can be checked separately from the large negative-channel calculations.

Section 3 of the source then obtains, with $c_0 = 37/27$, $q = c_0/(2+\lambda)$, $c = c_0 + 2\lambda q$,

$$0 \le \mathbb{E}J \le \mu - \lambda q\nu, \quad c\nu \le \eta \le \sqrt{\mu\nu}, \quad \mu^2/s \le \nu \le \mu/c^2.$$

Projection onto a sum of branch posterior means and a correlation increment bound imply

$$3H/\alpha \ge \frac{3\mu^2}{2(1+2\mu/3)} - \frac{\lambda}{\sqrt{2}}\mu\nu.$$

The ensuing scalar contradiction is split at $\lambda = 3/10$ and $1/\sqrt{2}$, using survival in the last Poisson interval. The report does not replace this analysis by M437's four-state Lyapunov cone. In particular, a negative edge changes the sign of y; a positive-channel radial statement cannot simply be evaluated at negative λ.

3.2 Negative equality: the exact pair obligations

At $\lambda = -a$, $0 < a \le 1/2$, $d = a^{-2}$, the source claims every nontrivial symmetric fixed point has $\mu > 1/20$. Its inputs are the current posterior invariants (X, y_0) and the next edge invariants $(z, B_0) = (a^2 w, -a^3 Y)$. Define tilted, six-relative-permutation pair excesses d_x, d_e, d_n of x, y, x^2, and

$$W = y_0 z + X B_0, \quad S_A = X^2 z, \quad S_B = X z^2, \quad S = S_A + S_B.$$

The exact seven inequalities needed are

$$\begin{aligned} d_n &\le \tfrac{13}{2} Xz, & -\tfrac{79}{250} Xz \le d_e &\le 11 Xz, \\ d_x &\le -Xz - 2W + 2y_0 B_0 + \tfrac{51}{20} S, & & \\ 6Xz - 6W + y_0 B_0 - \tfrac{19}{2} S \le d_e &\le 6Xz - 6W + y_0 B_0 + S, & & \\ \tfrac{5}{2} d_n - 2d_e &\le -2Xz + 22W - 7y_0 B_0 + 26 S_A + \tfrac{111}{2} S_B. & & \end{aligned} \tag{3.2}$$

They are not pointwise in a single orientation. At $a = 1/2$, sorted positive unnormalized inputs are parametrized as

$$p = (6s + 3t + 2, 3t + 2, 2), \quad q = (6u + 3v + 2, 3v + 2, 2), \qquad s, t, u, v \ge 0.$$

Put $P = \sum p_i$, $Q = \sum q_i$, $e_i = Q - q_i$, $E = 2Q$. In each of the two reflection classes, rotate $\widetilde{e}$, form $r_i^{(j)} = p_i \widetilde{e}_{i+j}$ and $T_j = \sum_i r_i^{(j)}$. The cyclic tilted weight is $T_j/(PE)$, and the cleared denominator is

$$D = PE(T_0 T_1 T_2)^3 P^4 E^4 > 0. \tag{3.3}$$

The reflection correction is

$$I_{\mathrm{odd}} = \frac{27 \operatorname{odd}(p) \operatorname{odd}(\widetilde{e})}{64 P^3 E^3}, \quad \operatorname{odd}(p) = 4(p_1 - p_2)(p_0 - p_1)(p_0 - p_2).$$

It changes sign between reflection classes. The denominators *also change*; cancellation is valid after dividing each class by its own D, not before. The program must show coefficientwise nonnegativity of seven degree-28 polynomials per reflection. Source-stated nonzero counts are $13672, 13672, 13674, 13625, 13625, 13625, 13625$ in either class. This claim is not established by sampling the four nonnegative parameters.

For $a < 1/2$, apply the coloring result to the convexly attenuated child $1 + 2av$. Boundary inputs follow by continuity of the complete tilted summands, not positivity of every T_j at the endpoint. The bootstrap uses partial regular products or observed Poisson mixtures, not the false comparison of every fixed count with the whole Poisson mixture. It first bounds $\nu/\mu^2 \le 4.94$, then proves $\eta \ge 0$, keeps $K_j = (5/2)\nu_j - 2\eta_j$ together, and obtains the final source excess upper bound

$$-\frac{2321911}{30000000}\,\alpha\mu^2 < 0 \quad (0 < \mu \le 1/20). \tag{3.4}$$

At a fixed point the actual averaged x excess is zero. All mixed moments factor only under the product marginal law, followed by the appropriate tilt already built into (3.2). The rational margin alone does not verify the fourteen polynomial premises.

3.3 Weighted-logarithm exclusion and interval coverage

Away from coloring set $G = (m_0 m_1 m_2)^{1/3}$, $T = 1 - G$, $W = G\langle u^2\rangle$, $Y = G\langle u^3\rangle$, and $\widehat{T} = dT_e$, etc. These extend to zero coordinates as $T = 1$, $W = Y = 0$. At a nontrivial fixed point, $A = \mathbb{E}G_e$, $z = d(1 - A)$, $h = (1 - \Phi(A))/(d(1 - A))$, $1 - \xi = \Phi'(A)/d$, with $\Phi(t) = \mathbb{E}t^D$, give

$$\mathbb{E}T = h\mathbb{E}\widehat{T}, \quad \mathbb{E}W = (1 - \xi)\mathbb{E}\widehat{W}, \quad \mathbb{E}Y = (1 - \xi)\mathbb{E}\widehat{Y}, \quad \mathbb{E}\widehat{T} = z.$$

Symmetry kills the weighted first log moments, so the cross terms in products of two or three centered logs vanish. The source uses $h \le 1 - \xi/2 - H\xi^2 - H\xi^3/2$ and $z \ge \xi + U\xi^2$ with $(H, U) = (1/16, 15/32)$, or $(1/12, 1/2)$ for the last Poisson band. The gap implies $\xi > .03$.

The zero-expectation combination is relaxed downward into $P(\xi) = \sum_{i=0}^{3} D_i \xi^i$. The sign of this relaxation matters: a strictly positive pointwise P contradicts $\mathbb{E}P \le 0$. The parameter choices are

a	$1000c_0$	$1000c_1$	$1000s_1$	1000δ	H	U
$[0, .190]$	-268	-60	-55	130	1/16	15/32
$[.190, .448]$	-269	67	47	55	1/16	15/32
$[.448, .477]$	-271	58	49	50	1/12	1/2

where $s_0 = c_0 + 1/3$, $c = c_0 + c_1\xi$, $s = s_0 + s_1\xi$. The checked coefficients are

$$\begin{aligned} D_0 &= T - \widehat{T} + c_0(W - \widehat{W}) + s_0(Y - \widehat{Y}), \\ D_1 &= c_1(W - \widehat{W}) + s_1(Y - \widehat{Y}) + c_0\widehat{W} + s_0\widehat{Y} + (1/2 - \delta)\widehat{T}, \\ D_2 &= c_1\widehat{W} + s_1\widehat{Y} + H\widehat{T} + \delta, \qquad D_3 = U\delta + (H/2)\widehat{T}. \end{aligned}$$

The sorted simplex parametrization is $p = (1, q^{18}, r^{18})/(1 + q^{18} + r^{18})$, $0 \le r \le q \le 1$. It gives continuously differentiable boundary functions $q^6(\log q)^j$, $j \le 3$. Edge expressions use the divided logarithm $\mathcal{L}(t) = -\log(1 - t)/t$, continuously extended by 1 at zero, avoiding division by a at $a = 0$.

The interval arithmetic uses scale 2^{40} and outward signed integer division, analytic tails for logarithm/divided-logarithm approximations, and checked wide products before narrowing. There are 4800 initial boxes per band. Twelve closed ξ intervals from .03 through 1 produce 48 cubic Bernstein coefficients, stored in 37 rows after shared endpoints are identified. Acceptance requires all rows positive either directly or by a midpoint-minus-gradient-radius enclosure. Subdivision must preserve closed coverage; depth or grid-resolution exhaustion is failure. The source records visit counts 414080, 756096, 1004426, plus separate undefined-behavior-sanitized runs. They remain source-recorded values unless a separate receipt explicitly establishes replay.

3.4 The 469-orbit certificate is a law, not a grid sample

The grid is $\mathcal{H}_H = \{h \in \mathbb{Z}^3_{\geq 0} : h_0 \geq h_1 \geq h_2, \sum h_i = H\}$, with $H = 72$, cardinality 469, and mass denominator $N = 10^{12}$. For integer raw posterior numerator r with $T = \sum r_i > 0$, the source sets $n_i = \lfloor Hr_i/T \rfloor$, $b_i = Hr_i - n_i T$, $k = H - \sum n_i \in \{0, 1, 2\}$. Its spread is n/H if $k = 0$, $(n + e_i)/H$ with weights b_i/T if $k = 1$, or $(n + \mathbf{1} - e_i)/H$ with weights $(T - b_i)/T$ if $k = 2$. The barycenter is r/T.

At rational magnitude $a = A/B$, $e_i(q) = (B + A)H - 3Aq_i$. The three product modes have raw coordinates and denominators

mode	r_i	D_{tab}
0	$p_i\, 3q_{\pi(i)}$	$6H^2$
1	$p_i e_{\pi(i)}(q)$	$6BH^2$
2	$e_i(p)e_{\pi(i)}(q)$	$18B^2H^2$

and tilted orientation weight T/D_{tab}. Aggregate exact spread numerators as $w_{p,q}(h)$; each table entry must sum to D_{tab}. Integer completion retains

$$L_h = \left\lfloor \frac{\sum_{p,q} m_p n_q w_{p,q}(h)}{N D_{\text{tab}}} \right\rfloor$$

units and puts all missing units at the pure orbit. This is a calibrated refinement by revealing the spin on removed outcomes. It cannot be replaced by normalizing rounded masses or choosing nearest grid points.

For regular coloring, the upper recursion merges edge–edge pairs and then plain–plain pairs. Poisson upper products keep the same child law while sequentially adding a branch; lower integer count weights and all omitted tail mass are completed by a pure experiment. The count may be forgotten only after proving that the labelled posterior is sufficient for the observed-count mixture.

There are 27 closed Poisson bins $[(800 + 3j)/200, (803 + 3j)/200]$, $0 \leq j \leq 26$, each covered by $d_+ = (803 + 3j)/200$ and $a_+ = A/10^6$ with A least such that $A^2(800 + 3j) \geq 200 \cdot 10^{12}$. The corner may be above criticality; the fixed-point gap is applied to the dominated critical target, not the corner. The union $[4, 881/200]$ overlaps the weighted-log range because

$$\frac{881}{200} - (1000/477)^2 = \frac{453049}{45505800} > 0.$$

The exact stopping test is

$$20 \sum_h m_h [H^2 - 3(h_0h_1 + h_1h_2 + h_2h_0)] < NH^2. \tag{3.5}$$

Source-stated stopping depths are 162 for the regular case; for the Poisson cases they are

$$104, 103, 103, 103, 102, 102, 102, 102, 102,$$
$$101, 101, 101, 101, 101, 101,$$
$$100, 100, 100, 100, 100, 100, 100, 99, 99, 99, 99, 99.$$

No value in this list is claimed as a run performed by this author. Here 469 counts stored permutation orbits, not labelled grid points: there are $\binom{74}{2} = 2701$ labelled points. Retaining their labels, not observing only the orbit index, is essential for the experiment interpretation.

4 Exact Spread Inflation and Its Limits

Lemma 4.1 (Barycentric refinement with its actual kernel)**.** *Let P be a calibrated posterior with law Q. Suppose $\alpha_p(z) \geq 0$, $\sum_z \alpha_p(z) = 1$, and $\sum_z \alpha_p(z) z = p$. Then the law $Q^+(z) = \int \alpha_p(z)\, \mathrm{d}Q(p)$ is calibrated and dominates Q.*

Proof. Given $S = i, P = p$ choose z with probability $\alpha_p(z) z_i / p_i$ when $p_i > 0$. The probability sums to one. If $p_i = 0$ the event has zero conditional probability and the kernel can be arbitrary. The joint law of (S, P, Z) is $Q(\,\mathrm{d}p)\alpha_p(z) z_i$, so its posterior given (P, Z) is z. The reverse kernel $Q(\,\mathrm{d}p)\alpha_p(z)/Q^+(z)$, for $Q^+(z) > 0$, is independent of S and returns the original conditional law. The construction explicitly distinguishes marginal spreading probabilities from conditional-on-spin refinement probabilities. □

Theorem 4.2 (Sharp second-moment cost of the source spread)**.** *Fix any integer $H \geq 1$ and any $p \in \Delta_2$. Apply the source barycentric rule with $n_i = \lfloor H p_i \rfloor$, $f_i = H p_i - n_i$, and $k = \sum_i f_i$. Let Z be the spread posterior. Then*

$$\mathbb{E}[x(Z) \mid p] - x(p) = \frac{3}{2H^2} \sum_{i=0}^{2} f_i(1 - f_i) \in [0, H^{-2}]. \tag{4.1}$$

The upper constant is attained. The rule preserves the labelled posterior barycenter, including faces and vertices. For a symmetric calibrated input law, integration gives a Blackwell upper law with the same bound on the increase in μ.

Proof. Every grid output coordinate equals either n_i/H or $(n_i + 1)/H$. In the $k = 1$ case its increment probability is f_i. In the $k = 2$ case that probability is $\sum_{j \neq i}(1 - f_j) = f_i$; for $k = 0$ all $f_i = 0$. Thus $\mathbb{E}Z_i = p_i$ and $\mathrm{Var}(Z_i \mid p) = f_i(1 - f_i)/H^2$. Substitute (2.2) to get the identity. If $k = 1$ or 2, Cauchy–Schwarz gives $\sum f_i^2 \geq k^2/3$, so $\sum f_i(1 - f_i) \leq k - k^2/3 = 2/3$. Equality holds when all three fractional parts equal $k/3$. Such a cell exists for every $H \geq 1$ using $k = 1$ and any nonnegative integer n summing to $H - 1$. Lemma 4.1 proves the order statement, including null coordinates. □

At $H = 72$ the sharp cost is $1/5184$. An explicit attainment point is $(73, 73, 70)/216$, whose three $H p_i$ have fractional part $1/3$. This is a rational identity, not an estimated variance or a measured reconstruction error.

Corollary 4.3 (Smooth convex observables)**.** *If a convex twice differentiable f has tangent-plane Hessian bounded above by L times the identity on the cell, then*

$$0 \leq \mathbb{E}f(Z) - f(p) \leq L/(3H^2).$$

The same holds by a continuous limiting argument at a boundary with this Hessian bound.

Proof. Taylor's integral remainder is at most $L\|Z - p\|_2^2/2$; the linear part vanishes in expectation. The preceding proof gives $\mathbb{E}\|Z - p\|_2^2 \leq 2/(3H^2)$. Jensen proves the lower bound. □

Entropy has unbounded Hessian near zero coordinates, so this corollary cannot be applied to it with an unstated global L. Nor may n/H^2 be asserted as a full depth-n error: subsequent nonlinear Bayes products change the input experiment. The bound is for each individual spread of the law actually presented to it.

5 A Less Informative Certified Completion

5.1 The extremal law for a truncated simplex

For $0 \le t \le 1/3$ write $c_i(t) = t\mathbf{1} + (1-3t)e_i$ and let C_t be their equally weighted labelled law. At $t = 1/3$ this is just the uniform posterior. Define $\Delta_2(t) = \{p \in \Delta_2 : p_i \ge t\}$.

Lemma 5.1 (Capped domination)**.** *Every symmetric posterior law supported in $\Delta_2(t)$ is dominated by C_t. Moreover $C_t \preceq C_0$, and $x(c_i(t)) = (1-3t)^2$. Within the family C_s, C_t is the least informative member that dominates every symmetric law supported in $\Delta_2(t)$.*

Proof. For $t < 1/3$ the unique barycentric weights in the three vertices are $\beta_i(p) = (p_i - t)/(1-3t)$. They are nonnegative, sum to one, and reconstruct p. By symmetry their mean is $1/3$, so Lemma 4.1 gives exactly C_t. At $t = 1/3$ the support is a singleton. A revealed spin can simulate every calibrated experiment, proving $C_t \preceq C_0$. The formula for x follows from $c_i(t) - \boldsymbol{u} = (1-3t)(e_i - \boldsymbol{u})$. If $s > t$, then $x(C_s) < x(C_t)$ for $t < 1/3$, so Jensen prevents C_s from dominating the admissible law C_t. If $s \le t$, the first part applied to C_t proves $C_t \preceq C_s$. This is minimality only in the specified one-parameter family, not a claim about all upper experiments. □

5.2 Stratum-aware fixed-denominator construction

Let an exact symmetric grid law have orbit masses $W_h \ge 0$, $\sum_h W_h = 1$. Its source completion retains $L_h = \lfloor NW_h \rfloor$ integer units. Put

$$\delta_h = W_h - L_h/N, \quad R_j = \sum_{h:h_2=j} \delta_h, \quad B_j = \lfloor NR_j \rfloor, \quad v_j = (H-2j, j, j), \quad 0 \le j \le \lfloor H/3 \rfloor. \tag{5.1}$$

All v_j are existing grid orbits. Retain L_h units, add B_j units to v_j, and put the remaining $N - \sum_h L_h - \sum_j B_j$ units at the pure orbit v_0. Call the resulting law $\mathcal{C}_{\mathrm{cap}}(W)$, and call the source pure completion $\mathcal{C}_{\mathrm{pure}}(W)$. Duplicate additions to an orbit are added, not overwritten.

Theorem 5.2 (Exact Blackwell sandwich and improvement ledger)**.** *For every $H, N \ge 1$ and every exact symmetric law W on $\mathcal{H}_H$, the construction above is a calibrated law with nonnegative integer masses summing to N, and*

$$W \preceq \mathcal{C}_{\mathrm{cap}}(W) \preceq \mathcal{C}_{\mathrm{pure}}(W). \tag{5.2}$$

The pointwise-in-input improvement is exactly

$$\mu(\mathcal{C}_{\mathrm{pure}}(W)) - \mu(\mathcal{C}_{\mathrm{cap}}(W)) = \sum_j \frac{B_j}{N}\left[1 - (1-3j/H)^2\right] \ge 0. \tag{5.3}$$

It is strictly positive if some $B_j > 0$ with $j > 0$. If J strata have $R_j > 0$, the mass requiring final pure completion is less than J/N (zero if $J = 0$).

Proof. Remove proportion δ_h/W_h from every labelled datum in orbit h, leaving the indicated retained mass; zero-mass orbits contribute nothing. This selection is permutation-invariant, so each residual stratum has uniform spin prior after normalization. In stratum j, with total mass $R_j > 0$, send the fraction $B_j/(NR_j)$ of the residual law through the capped refinement in Lemma 5.1, with $t = j/H$, and reveal the input spin for the remaining fraction. These fractions are chosen independently of the spin and original datum within that stratum. The normalized residual is supported in $\Delta_2(j/H)$, so the output of its capped part is exactly $C_{j/H}$. Forgetting flags and using posterior sufficiency produces the specified masses and proves the first order comparison.

For the second comparison, keep the source pure completion in flagged form: it retains the original L_h/N portion and reveals the spin on the residual portion. Conditional on that revealed spin and its residual stratum, simulate the capped experiment on fraction $B_j/(NR_j)$, keeping the rest pure. The stratum flag is independent of the spin under the removed mixture. This is a spin-independent kernel on the revealed datum, since that datum already includes the spin value. Finally forget flags. Alternatively reconstruct the flags from the pure-completed labelled posterior using its marginal conditional law; posterior sufficiency ensures this reverse kernel needs no unknown spin. Thus compression does not invalidate the garbling.

All masses are nonnegative and their total before the last addition is at most one, since $B_j/N \le R_j$. Each cap has $x = (1 - 3j/H)^2$, whereas the pure law has $x = 1$, giving (5.3). The final residual is $\sum_j (R_j - B_j/N)$, each positive-stratum summand in $[0, 1/N)$, which proves the last bound. For $j = H/3$ the residual law is uniform, so the singleton cap case is valid as well. □

For $H = 72$ at most 25 strata occur. Pure mass completion in the source pays for individual orbit deficits, which can involve 469 orbits. The new method uses exactly the same 469-orbit state space and N but preserves more of the posterior support information in the residual. This is a structural change in the upper experiment, not merely a smaller claimed rounding constant. Nevertheless $N = 10^{12}$ already makes that loss small; no substantial runtime consequence at those parameters follows from the bound alone.

Proposition 5.3 (One-step error ledger)**.** *Let W be the exact law after one barycentric spread of a symmetric law Q. With $K = |\mathcal{H}_H|$,*

$$\begin{aligned} 0 &\le \mu(\mathcal{C}_{\mathrm{cap}}(W)) - \mu(Q) \\ &\le H^{-2} + \sum_h \delta_h (1 - x(h/H)) - \sum_j (B_j/N)[1 - (1 - 3j/H)^2] \\ &< H^{-2} + (K-1)/N \qquad (K > 1). \end{aligned} \tag{5.4}$$

The middle expression is a sharp accounting identity for the completion contribution, plus the upper bound for spread.

Proof. Pure completion changes μ by exactly $\sum_h \delta_h(1 - x(h/H))$. Subtract (5.3) and use Theorem 4.2. Nonnegativity follows from the Blackwell order. The pure orbit has $x = 1$, so its deficit contributes zero; each of the remaining $K - 1$ deficits is less than $1/N$. If $K = 1$ the completion cost is zero, and the strict last inequality is replaced by the spread bound. □

Corollary 5.4 (Sharper source-grid rounding budget)**.** *For $H = 72$, every symmetric grid law satisfies*

$$0 \le \mu(\mathcal{C}_{\mathrm{pure}}(W)) - \mu(W) < \frac{49961}{144N}.$$

If W is obtained from Q by one source spread, and $S(W)$ denotes the exact saving in (5.3), then

$$0 \le \mu(\mathcal{C}_{\mathrm{cap}}(W)) - \mu(Q) < \frac{1}{5184} + \frac{49961}{144N} - S(W).$$

These bounds concern one operation on the given input law.

Proof. The pure-completion increase equals $\sum_h \delta_h(1 - x(h/72))$. Each $\delta_h < 1/N$, each cost is nonnegative, and at least one orbit has positive cost. Parametrize the 469 representatives by

$$h = (72 - 2c - t, c + t, c), \quad 0 \le c \le 24, \quad 0 \le t \le n_c := \lfloor (72 - 3c)/2 \rfloor.$$

Direct expansion gives $1 - x(h/72) = [144c - 3c^2 + (72 - 3c)t - t^2]/1728$. Summing powers of t yields the integer identity

$$\sum_{c=0}^{24} \left[(n_c + 1)(144c - 3c^2) + (72 - 3c)\frac{n_c(n_c + 1)}{2} - \frac{n_c(n_c + 1)(2n_c + 1)}{6} \right] = 599532.$$

Therefore the cost sum is $599532/1728 = 49961/144$. This finite identity was checked by both direct orbit enumeration and the displayed power-sum formula. The strict deficit inequalities prove the first bound; Theorem 4.2 and the exact saving prove the second. □

This sharpening was supplied by the independent reviewer and is integrated with attribution. It is not a propagated depth error or evidence for an earlier stopping depth.

5.3 An exact, strictly improving finite witness

Take $H = 6$, $N = 2$ and the symmetric orbit law with equal mass on

$$h^{(1)} = (4, 1, 1), \quad h^{(2)} = (3, 2, 1), \quad h^{(3)} = (2, 2, 2).$$

Each has $W_h = 1/3$, so every $L_h = 0$. The strata have $R_1 = 2/3$, $R_2 = 1/3$, hence $B_1 = 1$, $B_2 = 0$. The source produces the pure law, with $\mu = 1$. The improved rule produces half of the capped orbit $(4, 1, 1)/6$ and half pure, with $\mu = 5/8$. The exact original value is

$$\mu(W) = \tfrac{1}{3}(1/4 + 1/12 + 0) = 1/9.$$

Thus the certified improvement is $3/8$, and the full comparison is an experiment-order comparison, not just $1/9 \le 5/8 \le 1$. The deliberate coarse denominator makes the structural effect visible; it is not a benchmark for the source's $N = 10^{12}$ run.

5.4 Exact implementation and honest recursion scope

If $W_h = U_h/V$ share a common integer denominator with $\sum_h U_h = V$, compute $L_h = (NU_h) \,//\, V$ and integer residual numerators $r_h = NU_h - VL_h$. Then

$$B_j = \left(\sum_{h:h_2=j} r_h \right) // V.$$

No floating-point division is involved. For the source pair-combine accumulator $A_h = \sum m_p n_q w_{p,q}(h)$, the exact law is $W_h = A_h/(N^2 D_{\mathrm{tab}})$. Cancel the common factor *before* arithmetic: set $V_0 = ND_{\mathrm{tab}}$ and compute

$$L_h = A_h \,//\, V_0, \quad r_h = A_h - V_0 L_h, \qquad B_j = \left(\sum_{h:h_2=j} r_h \right) // V_0.$$

This avoids forming the unnecessary product NA_h, for which the following bound does not suffice. Group sums require wide integers. Under the source's actual parameters and modes only ($H = 72$, $N = 10^{12}$, mode 1 with reduced $B \le 10^6$, mode 2 at coloring with $B = 2$, and mode 0), its product bound below $2^{127} - 1$ also bounds all these nonnegative residual sums: $\sum_h r_h \le \sum_h A_h = N^2 D_{\mathrm{tab}}$.

This is not a bound for mode 2 at arbitrary rational A/B, arbitrary H, N, or the general finite-alphabet construction. Every changed implementation requires fresh bounds on all products, sums, indices, casts, and integer widths.

An upper recursion using this completion remains sound by induction: exact Bayes composition preserves experiment order, spread is refinement, and Theorem 5.2 is refinement. The map $W \mapsto \mathcal{C}_{\mathrm{cap}}(W)$ is not proved monotone in Blackwell order. Therefore (5.2) compares the two completions of the *same* exact incoming law; it does not prove that independently rerunning the new and old rounded iterations preserves their order at every depth. Earlier stopping requires actual runs. If a new run reaches (3.5), then, conditional on the source pair gap, its dominated critical target is non-reconstructible. There is no need to prove convergence of the rounded law.

For Poisson count tails, do not apply a positive cap merely because retained counts have positive-coordinate bounds. The omitted tail can involve arbitrarily many branches and its posteriors can approach a simplex vertex. Pure completion remains the default unless a common support constraint on the omitted conditional law is proved. The proposed stratum rule applies directly after an exact finite grid product or any other finite symmetric grid law whose masses are available, not to an unrepresented tail by guesswork.

6 Finite-Alphabet and Finite-Count Extensions

Proposition 6.1 (Alphabet-q capped completion)**.** *For a uniform prior on $q \geq 2$ states, replace $1/3$ by $1/q$, use $\Delta_{q-1}(t) = \{p : p_i \geq t\}$ and vertices $c_i(t) = t\mathbf{1} + (1 - qt)e_i$. Every permutation-invariant calibrated law on that truncated simplex is dominated by the uniform law on these vertices. Partitioning finite-grid residuals by their minimum coordinate and applying the same two-stage integer flooring gives the Blackwell sandwich (5.2). For the normalized quadratic information*

$$x_q(p) = \frac{q\sum_i p_i^2 - 1}{q - 1},$$

the exact saving is $\sum_j (B_j/N)[1 - (1 - qj/H)^2]$.

Proof. For $t < 1/q$, the barycentric probabilities $(p_i - t)/(1 - qt)$ sum to one and reconstruct p. Their averages are $1/q$ by symmetry. Apply the calibration/refinement proof with the conditional density qp_i, then use the residual-stratum construction verbatim. At $t = 1/q$ the simplex is a singleton. Finally $c_i(t) - \boldsymbol{u} = (1 - qt)(e_i - \boldsymbol{u})$, so $x_q(c_i(t)) = (1 - qt)^2$. This proves every assertion without using a reconstruction theorem for q states. □

Proposition 6.2 (A finite-count cap at an interior channel)**.** *Fix $-1/2 < \lambda < 1$, put $c_- = \min(1 - \lambda, 1 + 2\lambda) > 0$ and $c_+ = \max(1 - \lambda, 1 + 2\lambda)$. Every exact k-branch posterior from arbitrary child messages satisfies*

$$p_i^{\mathrm{out}} \geq t_k := \frac{c_-^k}{c_-^k + 2c_+^k}.$$

Hence a symmetric residual law of such exact posteriors can be completed to C_{t_k} instead of the pure law. After source grid spreading, it is safe to replace t_k by $\lfloor Ht_k \rfloor / H$.

Proof. In (2.5), $c_-^k \leq W_i \leq c_+^k$. For a fixed coordinate, $W_i/(W_i + \sum_{j \neq i} W_j) \geq c_-^k/(c_-^k + 2c_+^k)$. Every spread coordinate is at least $\lfloor Hp_i \rfloor / H \geq \lfloor Ht_k \rfloor / H$. Lemma 5.1 applies to any symmetric residual supported there. For $k = 0$, the exact law is uniform and $t_0 = 1/3$. □

At the coloring endpoint $c_- = 0$, the positive cap degenerates for $k > 0$. Nor does a finite-count formula certify an infinite Poisson tail. These extensions change a finite verification component, not the all-degree threshold itself. The q-state statement concerns probability laws and is independent of M437's special four-state inequalities.

Proposition 6.3 (A uniform finite Poisson weight cutoff)**.** *Let $4 \le d \le 881/200$, $N = 10^{12}$, and let nonnegative integers satisfy $w_0 \le Ne^{-d}$ and $w_n = \lfloor dw_{n-1}/n \rfloor$ for $n \ge 1$. Then $w_{27} = 0$, and all later weights vanish. This sharpens the source appendix's sufficient count bound of 48 for its prescribed range, without changing the omitted-count completion rule.*

Proof. Induction gives $w_n \le Ne^{-d}d^n/n!$. The positive exponential series implies

$$e^d \ge e^4 > \sum_{k=0}^{7} \frac{4^k}{k!} = \frac{16319}{315} > 50.$$

The exact integer inequality $10^{12}881^{27} < 50 \cdot 200^{27} \cdot 27!$ therefore gives $0 \le w_{27} < 1$. Integrality forces zero, and the recurrence propagates it. The integer comparison is recorded in the author check receipt, independently of the reviewer's calculation. No floating evaluation of an exponential is used. □

The reviewer supplied this continuous-parameter argument and separately checked the source's degree-41 directed initial-weight calculation for all 27 endpoints. The first zero is at count 26 in bins 0–21 and at 27 in bins 22–26. Thus the source count-loop denominators $200n$ need not exceed 5400 in these runs. The weight deficits before orbit-wise mixture rounding range from 29 to 110 integer units; these include both weight rounding and discarded tail, not just a tail probability. A valid initial lower weight remains an explicit premise of Proposition 6.3; no positive support cap is assigned to the omitted Poisson law.

7 Specific Private Ram Correspondences

7.1 Passages actually read

The following confidential originals, not their routing maps, were used. Paths are absolute and are also hashed in the provenance ledger.

1. `C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`: "What a Ram representation means in this report", equation `eq:moments`, and "A concrete polynomial Ram construction". These give $m = Ec$, synthesis by an inverse on the declared finite space, positive-weight Gram unisolvence, and the warning that an invertible coordinate change is not a new approximation theorem.

2. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume16/chapters/ch05d-clearance.tex`: the entire chapter, specifically `v16c05:bernstein`, `v16c05:conversionproof`, `v16c05:hull`, "Subdivision without introducing approximation", and the separate remainder/bias sections. Its algebraic basis conversion and continuous-domain enclosure are classical, with a precise private integration into a Ram workflow.

3. `C:/Users/mural/OneDrive/Documents/RamPatsCore50-TRs/TR-core-pat-16/TR-core-pat-16.tex`: "A probability model before moment computation", "One Joint Law Across Local Charts", "Common-polynomial contraction", "Validity, support, and reconstruction", `prop:psd`, `prop:gaussian`, `thm:remainder`, and `prop:rankloss`. The actual-law requirement

and covariance-versus-remainder distinction constrain the proposed computation. Gaussian formulas are not used for the non-Gaussian posterior recursion.

M437's report was read selectively for its law/Gram distinction and review boundaries. Its positive four-state polynomial cone, all-merges drift, and resulting rates are not premises for a three-state negative-channel proof.

7.2 Concrete moment realization of the interval certificate

For a fixed (q, r, a) box and one ξ interval $[l, h]$, let $t = (\xi - l)/(h - l)$ and write the cubic as $P = \sum_{j=0}^{3} \beta_j B_j^3(t)$. The reference law is *uniform Lebesgue probability on* $[0, 1]$, not the posterior law. Define Ram moments

$$z_j = \int_0^1 P(t) B_j^3(t)\, \mathrm{d}t, \quad z = E\beta, \quad E_{jk} = \binom{3}{j}\binom{3}{k}\frac{(j+k)!(6-j-k)!}{7!}. \tag{7.1}$$

Proposition 7.1 (Exact finite Ram cone)**.** *E is rational positive definite. Exact synthesis $\beta = E^{-1}z$ identifies the same positivity certificate as the original Bernstein test: $E^{-1}z > 0$ componentwise implies $P > 0$ throughout $[0, 1]$. If $\widehat{z}$ has rigorous errors $|z_j - \widehat{z}_j| \le \epsilon_j$, a sufficient robust test is*

$$(E^{-1}\widehat{z})_i > \sum_j |(E^{-1})_{ij}|\epsilon_j \quad \textit{for every } i.$$

If S is an exact Bernstein subdivision matrix, the child moment map is ESE^{-1}, using its own normalized coordinate on $[0, 1]$.

Proof. For a nonzero vector c, $c^T E c = \int (\sum c_j B_j^3)^2 > 0$, by polynomial independence and positive measure on every open interval. The beta integral gives the displayed entries. Rational invertibility follows. Nonnegative Bernstein functions sum to one, so a positive coefficient minimum bounds P below. The error estimate follows componentwise from the triangle inequality. Synthesis, subdivision, and reanalysis give ESE^{-1}; no physical-length Jacobian is missing because the child measure has explicitly been renormalized. Physical unnormalized integration would require that length factor. □

On $[l, h]$ the Bernstein rows multiplying (D_0, D_1, D_2, D_3) are

$$(1, l, l^2, l^3), \quad (1, (2l + h)/3, l(l + 2h)/3, l^2 h),$$

$$(1, (l + 2h)/3, h(h + 2l)/3, lh^2), \quad (1, h, h^2, h^3).$$

Thus the moment realization is tied to the exact source acceptance rows. The logarithmic dependence on (q, r, a) still needs the interval enclosures; a cubic moment transform does not make those functions polynomial in all variables.

7.3 The pair polynomial and posterior moments use different measures

For a cleared pair certificate $f(s, t, u, v)$ of total degree at most 28, one possible exact bank is the monomial basis indexed by $|\alpha| \le 28$, of dimension $\binom{32}{4} = 35960$, with reference probability density $e^{-(s+t+u+v)}$ on $\mathbb{R}^4_{\ge 0}$. Its Gram matrix is

$$E_{\alpha\beta} = \prod_{i=1}^{4} (\alpha_i + \beta_i)!.$$

All integrals are finite. Positivity of the density and polynomial independence prove positive definiteness exactly as above. Thus $E^{-1}z$ recovers the ordinary coefficients, whose nonnegativity proves $f \geq 0$ on the entire orthant. This is an explicit finite realization of TR250's definition. It is *not* an efficient proposal to invert a dense 35960-dimensional matrix; source sparse coefficient convolution is likely the more appropriate implementation absent benchmarks.

In contrast, actual posterior-law moments are $\mathbb{E}_Q\phi(P)$, and their Gram matrix is $\mathbb{E}_Q[\phi\phi^T]$. They are positive semidefinite because Q is a common proper law. Products of separate branch moments factor under $Q_1 \otimes Q_2$, but the output tilt has rational denominators. A finite moment list is not an exact closure of (2.6). In particular, polynomial moments do not determine an arbitrary rational posterior observable.

For a small negative control, take degree one on $[0,1]$, $B_0 = 1-t$, $B_1 = t$. Then $E = \frac{1}{6}\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and $E^{-1} = \left(\begin{smallmatrix} 4 & -2 \\ -2 & 4 \end{smallmatrix}\right)$. The polynomial $P(t) = t - 1/8$ has coefficients $(-1/8, 7/8)$ but moments $(5/48, 13/48)$, both positive. It is negative at zero. Hence positive moment entries, or a PSD Gram matrix, cannot replace the reconstructed-coefficient sign test. The private corpus explicitly warns against that inference.

The completion improvement in Section 5 uses these private principles in their proper roles: it preserves the actual probability law and its support before evaluating moments, keeps exact representation separate from conservative refinement, and records the information increase rather than hiding it inside a covariance. The proof itself is classical barycentric/Blackwell reasoning specialized to the source's finite experiment, not a newly discovered Ram transform identity.

8 Verification, Comparisons, and Remaining Work

8.1 Bounded verification contract

The author check program `_audit/M439-author-check.py` uses exact Python integers and `Fraction` arithmetic; its receipt is `_audit/M439-author-checks.json`. It passed 14620 spread fixtures with raw denominator at most 24 and $H \in \{1, 2, 3, 6, 72\}$, including zeros, as well as the separate sharp $H = 72$ witness. It passed 525 completion fixtures formed from all triples of the seven $H = 6$ orbits, three declared rational mass patterns, and $N \in \{1, 2, 3, 5, 11\}$. Both exact posterior-martingale couplings were checked in each case; 38 cases had a strictly positive saving. These are complete enumerations of the declared finite fixtures, not of all probability laws.

The same program counted 469 $H = 72$ orbits, verified the cubic Gram inverse by rational elimination, found determinant $3/224000$ and positive elimination pivots $1/7, 1/20, 3/100, 1/16$, and checked all four power-basis polynomials on each of the twelve actual source ξ intervals (48 coefficient-identity tests). It also checked the positive-moments/negative-polynomial counterexample. The first attempt stopped at the Gram step because SymPy was unavailable in the bundled runtime; the final successful program uses no SymPy and no package was installed. A finite test of a general proposition is supporting evidence; its proof is the argument above. Independent source-algebra and table checks, when cited below, are a different receipt, not tests performed by this author.

After review, the author program additionally verified the 2701 labelled grid points, the exact cost sum in Corollary 5.4 by two finite formulas, and the integer comparison and Taylor partial sum in Proposition 6.3. These additions do not launch any original threshold search.

8.2 Independent bounded audit integrated

The independent reader supplied `_audit/M439-independent-checks.py` and `_audit/M439-independent-evidence.txt`. The evidence records 14 symbolic identities, including the radial differential

polynomial and its substitutions, the weighted cubic coefficients, and Bernstein conversion; the radial sign and integration argument was also inspected on its positive-message domain. It records exact interval-tail constants, all 27 parameter-bin choices and directed Poisson weight constructions, and 1083 rational pair cases (all 19^2 pairs of $H = 12$ orbits at $a = 0, 1/4, 1/2$). All tested inequalities passed. These are not the fourteen degree-28 coefficient expansions or complete interval traversals.

The reader exhaustively enumerated 440391 stored pair rows at $H = 72$, $A/B = 1/2$, across the three table modes, covering 2642346 relative permutations. Checks included exact row mass, grid-corner membership, barycenters, and the spread cost. The triangular modes 0 and 2 and ordered mode 1 have different row conventions; their zero-normalizer counts were respectively 76, 0, and 2. This is a bounded Python reimplementation of table mathematics at coloring, not execution of the recovered C++ tables for all parameter bins, nor a sample of the large stopping run.

For four-child coloring, the independent depth-one law is pure with mass 7/8 and on the orbit $(1/2, 1/2, 0)$ with mass 1/8, giving $\mu_1 = 29/32$. At depth two, the source-style upper reimplementation has six pair orbits and 29 output orbits, with

$$\mu_2^{\text{upper}} = \frac{743474974896781}{864000000000000}, \qquad \mu_2^{\text{exact}} = \frac{297758309}{346030080},$$
$$\mu_2^{\text{upper}} - \mu_2^{\text{exact}} = \frac{776197067831}{152064000000000000} > 0.$$

The exact comparison enumerates $6^4 = 1296$ labelled outcomes, 108 with zero likelihood, with total tilted mass one. Neither moment is below 1/20. This is a substantive shallow verification of domination, not a reproduction of the regular stopping depth 162. The reader did not review this report's new completion proofs; those received the parent's separate targeted mathematical review.

8.3 Recovered code and unreplayed dependencies

The source package's local mathematical appendices were present, but `verification/pair.py`, `interval.cpp`, `grid.cpp`, and `reproduce.py` were absent from the local selected-source snapshot. They were subsequently recovered from the public repository at the exact pinned commit, saved only under `_audit/M439-author-source-*`, and read in full. The first three byte hashes match the appendix exactly:

`pair.py`: `ab775c3a63fb1d2713bfe058af2dece4ceda99627c17ce18f51f817b69410bba`
`interval.cpp`: `76d1443b1194597ee77749e8030c825f2b9aae46b5c172d9eccf0745078c28e3`
`grid.cpp`: `03c92321efedd0db53b1e91372918dfcd68d459db0223cb160c39df3bd645283`.

The recovered runner hash is recorded in the read-scope ledger; the appendix does not prescribe a runner hash. Static reading confirms that `pair.py` constructs both reflection classes and checks minimum polynomial coefficients, `interval.cpp` implements the outward intervals and subdividing acceptance test described above, and `grid.cpp` constructs full pair tables before its upper-law iteration. In the latter, table mass is asserted for every pair, the pure completion is explicit, zero normalizers are skipped, and only progress diagnostics use floating-point conversions. These source-to-formula observations do not establish completion of their computations or absence of every implementation defect.

A source hash establishes identity, not correctness. The source's stated complete replay is 37 passing subprocess records, consisting of 32 mathematical executions and five environment/compiler checks, plus a separate three-case interval sanitizer protocol. Neither a successful local unit test nor a symbolic radial identity establishes that full replay. No original pair polynomial execution, interval traversal, sanitizer run, or full grid threshold run was performed by this author.

The source machine model requires assertions enabled, signed `__int128`, plain `int` of at least 32 bits, exact negative floors, and overflow checks before narrowing. The grid denominators, nonnegative accumulator bounds, count-tail construction, and strict comparison (3.5) must all survive any implementation change. Poisson degree-41 alternating Taylor lower bounds and downward count weights are one-sided evidence, not rounded floating-point approximations.

After recovery, the independent reader also verified the three prescribed hashes and read the mathematical programs. Its grid width analysis is confined to the original valid modes and parameters, as in Section 5. It found no overflow error in those prescribed grid calls. A residual interval-code qualification remains: the signed `visits++` counter is unchecked, and a depth-80 subdivision limit does not itself bound the total visit count below the integer limit. The source appeals to completed sanitizer executions, which neither author nor reviewer reproduced. A future implementation could check the counter before incrementing it. The interval program's argument protocol is also restricted to $0, 1, 2$; static inspection is not a generic input-safety guarantee. No original program was modified.

8.4 Current public primary comparisons

The public literature was searched by public titles and mathematical topics on October 7, 2026; private passages were not uploaded or used as queries. Bhatnagar–Maneva's finite surveys already supply the core idea of certified posterior-law discretization, including refinement rather than plain sampling [2]. Sly's three-state work establishes large-degree sharpness and illustrates the need to combine a local expansion with a global argument [3]. Mossel–Sly–Sohn establish large-degree three/four-community results and a tree-to-graph transfer, not the all-degree theorem independently asserted here [4]. Formentin–Kulske's symmetrized entropy and Gu–Polyanskiy's nonlinear information contraction are genuine predecessors of the entropy route [5, 6]. Gu's June 2026 binary-hypertree work combines robust non-reconstruction and rigorous population dynamics and discusses a Potts extension [7]; binary less-noisy tools do not automatically prove the present three-state negative-channel inequalities.

The realized addition relative to the pinned manuscript is the proved residual-stratum completion, its exact improvement ledger and finite-alphabet extension, together with the sharp source-spread budget. It does not claim to originate surveys, convex order, Bernstein enclosures, Gram matrices, or information monotonicity. No historical novelty search sufficient for a priority claim has been performed.

8.5 Pros, cons, and concrete next steps

The capped rule keeps the exact integer mass representation and existing orbit grid, supplies an explicit garbling proof, and can preserve substantial information at coarse mass resolution. Its exact saving can be read off from residual buckets without evaluating logarithms or solving an optimization problem. The sharp spread identity separates spatial discretization loss from mass rounding loss.

Costs and limits are material. Additional bucket accumulation and wide-integer accounting are required. At the source's large N, mass-rounding improvement may be negligible compared with spatial spreading. A more informative upper law remains conservative; it is not an approximation with an automatically propagated depth-error bound. No monotonicity of the rounded map, termination rate, or reduced number of iterations is proved. Dense Ram Gram coordinates can destroy sparse structure and introduce conditioning issues; exact algebra avoids roundoff but not coefficient growth.

The next meaningful experiments are: replay the original certificates with verified identities and assertions; compare the two completions on identical incoming 469-orbit laws; record the exact sum in (5.3); then independently run modified recursions under a fixed resource budget. Any claimed speedup must include table construction, arithmetic, and memory. For tails, derive a genuine common support bound or leave pure completion intact. For approximate moments, use the robust inverse-Gram inequality with outward errors and retain the logarithmic remainder budget. For a larger negative fixed-point gap, rerun the entire bootstrap with valid pair premises and all signs, rather than optimizing only its final numerical line.

9 Provenance and Review Record

The source read scope comprises `build/main.tex`, `build/preamble.tex`, all files in `build/sections/`, `build/references.bib`, and both READMEs. The exact input hashes, individual read ranges for private files, public URLs, commands, check outputs, and review status are in `_audit/M439-author-*`. The private TR250 and pattern-report passages were read selectively as specified in Section 7; no whole-corpus reading is asserted.

The parent reported that native source opening was queued and the built-in compiler returned `compile-failed` with the platform error "Unable to find standard directories for platform" followed by a newline. This actual response is preserved in `_audit/M439-author-native.json`; no native compilation success is claimed. The installed MiKTeX disk export used the existing `_tools/compile_report.py` and bundled Python, without installation. Round 1 exposed one overfull stopping-depth list, subsequently split into three rows. The final second round's compilation, full-page contact inspection, and detailed samples are recorded in the build and visual-QA receipts; disk export is a separate result from native compilation.

The parent read the report and approved the local spread, cap, sandwich, ledger and finite-alphabet proofs, while requiring the original-mode restriction on wide-integer bounds. That restriction and the independent reader's finite bounds, bounded checks, static-code qualifications, and unreplayed obligations are incorporated above. These are internal targeted reviews, not external peer review or a full source-theorem audit. The final manifest records report, PDF, check, review-evidence and input hashes. The realized contribution is C3: a proved local refinement with explicit witnesses and extensions, not a completed new global threshold or demonstrated computational acceleration. No catalogue or delivery record was modified by this author.

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[8] Confidential Ram foundation TR250 revision. *Compatible Variational Ram Panels*. Actual finite moment and polynomial passages identified by absolute path in Section 7; no public upload.

[9] Confidential Ram Calculus Books, Volume 16, chapter `ch05d-clearance.tex`. Bernstein conversion, hull, subdivision and remainder passages, as identified in Section 7.

[10] Confidential `TR-core-pat-16.tex`. Proper law, joint moments, support validity, remainders and rank loss, as identified in Section 7. No patent-validity or priority assessment.